

SMALL-ORDER DERIVATIVES OF RANDOM POLYNOMIALS UNDER LOGARITHMIC NON-CONCENTRATION

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ABSTRACT. Let $X_1, \dots, X_n$ be i.i.d. complex random variables with law μ , and set $P_n(z) = (z - X_1) \cdots (z - X_n)$. We study the empirical zero measure of the k_n -th derivative $P_n^{(k_n)}$ in the small-order regime $k_n = o(n)$. We prove convergence in probability to μ whenever μ is discrete or contains a positive compact logarithmically non-concentrated component. Under the stronger assumption that this component satisfies an L^p logarithmic non-concentration condition for some $p > 2$, the convergence holds almost surely. As a consequence, almost-sure convergence holds for every $k_n = o(n)$ when μ has a positive compact Frostman component, and in particular for the Doeblin and dimension-nondegenerate classes considered by Angst, Nguyen and Poly.

1. INTRODUCTION

Let $X_1, \dots, X_n$ be i.i.d. random variables with values in $\mathbb{C}$. Let μ be the distribution of X_1 . Consider a random polynomial

$$P_n(z) := \prod_{j=1}^n (z - X_j).$$

Let $W_1^{(n,k)}, \dots, W_{n-k}^{(n,k)}$ be the zeros of $P_n^{(k)}$, counted with multiplicity, and define

$$\mu_n := \frac{1}{n} \sum_{j=1}^n \delta_{X_j}, \quad \mu_{n,k} := \frac{1}{n-k} \sum_{\ell=1}^{n-k} \delta_{W_\ell^{(n,k)}}.$$

We want to know whether we have

$$\mu_{n,k_n} \Rightarrow \mu$$

almost surely when $k_n = o(n)$. For $k = 1$, Kabluchko proved convergence in probability for arbitrary μ [Kab15]; Angst, Malicet and Poly later upgraded the first-derivative result to almost sure convergence [AMP24]. For higher derivatives, Byun, Lee and Reddy proved convergence in probability for fixed k [BLR22]; Michelen and Vu proved convergence in probability up to $k \leq \log n / (5 \log \log n)$ and almost sure convergence for fixed k [MV24b, MV24a]. The recent work of Angst, Nguyen and Poly proves almost sure convergence for all $k = o(n)$ when μ is discrete, and for $k = o(n / \log n)$ when μ is dimension-nondegenerate; their Theorem 1.4 also treats Doeblin measures up to $o(n / \log n)$ and a generic Cauchy-transform class up to $o(\sqrt{n})$ in probability [ANP26].

1.1. Notions. For $x > 0$, define

$$\log^- x := \max\{0, -\log x\}, \quad \log^+ x := \max\{0, \log x\}.$$

Definition 1.1 (Positive compact logarithmically non-concentrated component). Let $p \geq 1$. We say that μ has a *positive compact L^p -logarithmically non-concentrated component* if there exists a compactly supported finite measure $0 < \nu \leq \mu$ such that, with

$$\bar{\nu} := \frac{\nu}{\nu(\mathbb{C})},$$

we have

$$\sup_{w \in \mathbb{C}} \int_{\mathbb{C}} (\log^- |x - w|)^p d\bar{\nu}(x) < \infty.$$

When $p = 1$, we simply say that μ has a positive compact logarithmically non-concentrated component.

A more geometric, easier-to-check sufficient condition is the *Dini-type concentration condition*.

Definition 1.2 (Dini-regular component). We say that a probability measure on $\mathbb{C}$, say σ , satisfies the *Dini- p condition* if

$$\int_0^1 \left(\log \frac{1}{r} \right)^{p-1} \frac{\sup_{w \in \mathbb{C}} \sigma(D(w, r))}{r} dr < \infty.$$

We say that μ has a *positive compact Dini- p component* if there exists a compactly supported finite measure $0 < \nu \leq \mu$ such that $\bar{\nu}$ satisfies the Dini- p condition.

The most convenient checkable stronger condition is the following.

Definition 1.3 (Frostman component). We say that μ has a *positive compact Frostman component* if there exists a nonzero finite measure $\nu \leq \mu$ with compact support and constants $C < \infty$, $s > 0$ such that

$$\nu(D(w, r)) \leq Cr^s \quad \text{for every } w \in \mathbb{C}, 0 < r \leq 1.$$

1.2. Main results.

Theorem 1.4 (Main theorem). Let $k_n \in \{0, 1, \dots, n-1\}$ satisfy $k_n = o(n)$. Assume that either (i) μ is discrete, or (ii) μ has a positive compact logarithmically non-concentrated component. Then

$$\mu_{n, k_n} \Rightarrow \mu \text{ in probability.}$$

Under a similar proof scheme and a stronger L^p -logarithmic condition we obtain the almost-sure version

Theorem 1.5. Let $k_n \in \{0, 1, \dots, n-1\}$ satisfy $k_n = o(n)$. Assume that either (i) μ is discrete, or (ii) μ has a positive compact L^p -logarithmically non-concentrated component for some $p > 2$. Then

$$\mu_{n, k_n} \Rightarrow \mu \text{ almost surely.}$$

For a Frostman component ν , the normalized measure $\bar{\nu}$ satisfies the Dini- p condition for every finite p . By the layer-cake formula, the Dini- p condition implies the L^p logarithmic non-concentration condition. Therefore we have the following corollary.

Corollary 1.6 (Frostman, dimension-nondegenerate, and Doeblin components). Let $k_n = o(n)$. Assume that either (i) μ is discrete, or (ii) μ has a positive compact Frostman component. Then

$$\mu_{n, k_n} \Rightarrow \mu \text{ almost surely.}$$

In particular, the conclusion holds for the Doeblin class of [ANP26]. It also holds for every dimension-nondegenerate measure in the sense of [ANP26].

Remark 1.7. The discrete part of Theorem 1.5 is already contained in [ANP26, Theorem 1.3]. The new contribution is the non-discrete branch. Under a Frostman, Doeblin, or dimension-nondegenerate component, Corollary 1.6 gives almost sure convergence for the full range $k = o(n)$, whereas [ANP26, Theorems 1.3 and 1.4] give $k = o(n/\log n)$ for the non-discrete classes.

2. PRELIMINARIES

The following identity is standard.

Lemma 2.1 (Green's identity). For every $\phi \in C_c^\infty(\mathbb{C})$ and every $a \in \mathbb{C}$,

$$\phi(a) = \frac{1}{2\pi} \int_{\mathbb{C}} \Delta \phi(z) \log |z - a| d^2 z.$$

We use the standard logarithmic-potential reduction as in [Kab15], [BLR22], and [ANP26]. For $z \notin \{X_1, \dots, X_n\}$, differentiating the product gives

$$P_n^{(k)}(z) = k! \sum_{1 \leq i_1 < \dots < i_k \leq n} \prod_{j \notin \{i_1, \dots, i_k\}} (z - X_j),$$

therefore

$$\frac{P_n^{(k)}(z)}{(n)_k P_n(z)} = \binom{n}{k}^{-1} e_k \left(\frac{1}{z - X_1}, \dots, \frac{1}{z - X_n} \right),$$

where

$$e_k(y_1, \dots, y_n) := \sum_{1 \leq i_1 < \dots < i_k \leq n} y_{i_1} \cdots y_{i_k}$$

is the elementary symmetric polynomial and $(n)_k := n(n-1) \cdots (n-k+1)$. Set

$$R_{n,k}(z) := \frac{P_n^{(k)}(z)}{(n)_k P_n(z)} \quad \text{for } k \geq 1, \quad \text{and} \quad R_{n,0}(z) \equiv 1.$$

Then since

$$P_n^{(k)}(z) = (n)_k \prod_{j=1}^{n-k} (z - W_j^{n,k}),$$

we have the following potential identity.

Lemma 2.2. *For every $\phi \in C_c^\infty(\mathbb{C})$ and every $0 \leq k < n$,*

$$\frac{1}{n} \sum_{\ell=1}^{n-k} \phi(W_\ell^{n,k}) - \frac{1}{n} \sum_{j=1}^n \phi(X_j) = \frac{1}{2\pi n} \int_{\mathbb{C}} \Delta \phi(z) \log |R_{n,k}(z)| d^2 z.$$

The positive part of the logarithm is controlled deterministically with no assumption on μ .

Lemma 2.3. *For every compact set $K \subset \mathbb{C}$, there is a constant $C_K < \infty$ such that almost surely for all n and all $0 \leq k < n$,*

$$\int_K \frac{1}{n} \log^+ |R_{n,k}(z)| d^2 z \leq C_K \frac{k}{n}.$$

In particular, if $k_n = o(n)$, then the integral is $o(1)$.

Proof. $k = 0$ is trivial. For $k \geq 1$, Maclaurin's inequality gives

$$|R_{n,k}(z)| \leq \left(\frac{1}{n} \sum_{j=1}^n \frac{1}{|z - X_j|} \right)^k$$

for Lebesgue-a.e. z . Since $\log^+ u \leq u$ for $u \geq 0$,

$$\log^+ |R_{n,k}(z)| \leq k \frac{1}{n} \sum_{j=1}^n \frac{1}{|z - X_j|}.$$

We claim that $\sup_{w \in \mathbb{C}} \int_K \frac{1}{|z-w|} d^2 z < \infty$. Indeed, assume $K \subset B(0, R)$ and $|w| \leq 2R$, then $K - w \subset B(0, 3R)$; if $|w| > 2R$, then $|z - w| \geq |w| - R > R$ on K by triangular inequality. $\square$

Lemma 2.4. *Let $p \geq 1$, and let μ be a compactly supported probability measure on $\mathbb{C}$ with $\sup_{w \in \mathbb{C}} \int_{\mathbb{C}} (\log^- |x - w|)^p d\mu(x) < \infty$. Then for every compact $K \subset \mathbb{C}$,*

$$\sup_{z \in K} \sup_{w \in \mathbb{C}} \int \left(\log^- \left| \frac{1}{z - x} - w \right| \right)^p d\mu(x) < \infty.$$

In particular, μ is non-atomic.

Proof. If μ has an atom, say at a , then

$$\int (\log^- |x - a|)^p d\mu(x) = \infty$$

contradicting to the hypothesis. So μ is non-atomic.

Choose $R \geq 1$ so that $|z - x| \leq R$ for every $z \in K$ and $x \in \text{supp } \mu$. Fix $z \in K$. If $w = 0$,

$$\log^- \left| \frac{1}{z - x} \right| = \log^+ |z - x| \leq \log R.$$

If $w \neq 0$, set $a = z - 1/w$, then for μ -a.e. x ,

$$\left| \frac{1}{z-x} - w \right| = \frac{|x-a|}{|z-x||z-a|} \geq \frac{|x-a|}{|z-x|(|z-x|+|x-a|)} \geq \frac{|x-a|}{R(R+|x-a|)}.$$

This implies

$$\log^- \left| \frac{1}{z-x} - w \right| \leq C_R + \log^- |x-a|.$$

Raising to the power p and integrating gives

$$\int \left(\log^- \left| \frac{1}{z-x} - w \right| \right)^p d\mu(x) \leq C_{p,R} \left(1 + \int (\log^- |x-a|)^p d\mu(x) \right) < \infty.$$

□

3. SETUPS

This section contains the main probabilistic input. The first estimate is the L^1 logarithmic recursion, which gives convergence in probability. The second is the L^p version, which gives summability and hence almost sure convergence.

Lemma 3.1. *Let $Y_1, \dots, Y_m$ be i.i.d. non-atomic complex random variables such that*

$$L := \sup_{w \in \mathbb{C}} \mathbb{E} \log^- |Y_1 - w| < \infty.$$

Fix $0 \leq r \leq m$ and complex numbers $b_0, b_1, \dots, b_r$ with $b_0 = 1$. Define

$$F_{m,r} := \sum_{j=0}^r b_j e_{r-j}(Y_1, \dots, Y_m).$$

Then $F_{m,r} \neq 0$ almost surely and

$$\mathbb{E} \log^- |F_{m,r}| \leq rL.$$

Consequently,

$$\mathbb{E} \log^- \left| \binom{n}{r}^{-1} F_{m,r} \right| \leq \log \binom{n}{r} + rL$$

whenever $r \leq n$.

Proof. $r = 0$ is immediate since $F_{m,0} = 1$. For $r \geq 1$, use the elementary recursion

$$e_k(Y_1, \dots, Y_m) = e_k(Y_1, \dots, Y_{m-1}) + Y_m e_{k-1}(Y_1, \dots, Y_{m-1}).$$

Thus

$$F_{m,r} = F_{m-1,r} + Y_m F_{m-1,r-1}.$$

where $F_{m-1,r}$ and $F_{m-1,r-1}$ are defined with the same leading coefficient $b_0 = 1$ and the relevant initial segment of coefficients. By induction, $F_{m-1,r-1} \neq 0$ almost surely. And since after conditioning on $Y_1, \dots, Y_{m-1}$, the quantities $F_{m-1,r}$ and $F_{m-1,r-1}$ are fixed, the above identity shows that $F_{m,r} = 0$ can occur for at most one value of Y_m . Since Y_m is non-atomic, it has conditional probability zero. hence $F_{m,r}$ almost surely. Moreover,

$$F_{m,r} = F_{m-1,r-1}(Y_m + C_{m,r})$$

where

$$C_{m,r} := \frac{F_{m-1,r}}{F_{m-1,r-1}} = \frac{\sum_{j=0}^r b_j e_{r-j}(Y_1, \dots, Y_{m-1})}{\sum_{j=0}^{r-1} b_j e_{r-1-j}(Y_1, \dots, Y_{m-1})}.$$

Hence

$$\log^- |F_{m,r}| \leq \log^- |F_{m-1,r-1}| + \log^- |Y_m + C_{m,r}|.$$

Conditioning on $Y_1, \dots, Y_m$ gives the recursion

$$\mathbb{E} \log^- |F_{m,r}| \leq \mathbb{E} \log^- |F_{m-1,r-1}| + L.$$

Iterating r times gives the result. □

The following proposition is the integrated version needed for local logarithmic potentials.

Proposition 3.2. *Let μ be a compactly supported probability measure with $\sup_{w \in \mathbb{C}} \int_{\mathbb{C}} \log^- |x-w| d\mu(x) < \infty$. Let $K \subset \mathbb{C}$ be compact. Let $Z_1, \dots, Z_m$ be i.i.d. with law μ , and let $0 \leq r \leq m$. Let $\mathcal{G}$ be a σ -algebra independent of the Z_i , and let $b_0(z), \dots, b_r(z)$ be $\mathcal{G}$ -measurable functions of z , finite for Lebesgue-a.e. z , with $b_0(z) = 1$. Set*

$$F_{m,r}(z) := \sum_{j=0}^r b_j(z) e_{r-j} \left(\frac{1}{z-Z_1}, \dots, \frac{1}{z-Z_m} \right).$$

Then there is a constant $C_{K,\mu} < \infty$ such that

$$\mathbb{E} \left[\int_K \log^- |F_{m,r}(z)| d^2 z \mid \mathcal{G} \right] \leq C_{K,\mu} r \quad \text{almost surely.}$$

Proof. Since the coefficient functions are finite for Lebesgue-a.e. z , we fix such z . Then apply Lemma 3.1 to

$$Y_i(z) = \frac{1}{z-Z_i}.$$

By Lemma 2.4,

$$\sup_{z \in K} \sup_{w \in \mathbb{C}} \mathbb{E} \log^- |Y_i(z) - w| < \infty.$$

Integrating the resulting conditional expectation over K proves the proposition. $\square$

For the almost surely convergence, we need a summable tail bound. We will use the following martingale moment inequality.

Lemma 3.3. [HH80, Theorem 2.11.] *Let $p > 2$, and let $(D_j, \mathcal{F}_j)_{j=1}^r$ be a real-valued martingale difference sequence with $D_j \in L^p$. Then there exists a constant $C_p < \infty$ such that*

$$\mathbb{E} \left| \sum_{j=1}^r D_j \right|^p \leq C_p \left(\mathbb{E} \left(\sum_{j=1}^r \mathbb{E}[D_j^2 \mid \mathcal{F}_{j-1}] \right)^{p/2} + \sum_{j=1}^r \mathbb{E}|D_j|^p \right).$$

Proposition 3.4. *Let $p > 2$. Let μ be a compactly supported probability measure with $\sup_{w \in \mathbb{C}} \int_{\mathbb{C}} (\log^- |x-w|)^p d\mu(x) < \infty$. Let $K \subset \mathbb{C}$ be compact. In the notation of Proposition 3.2, there are constants $C_{K,\mu}, C_{p,K,\mu} < \infty$ such that for all $t > 0$,*

$$\mathbb{P} \left(\int_K \log^- |F_{m,r}(z)| d^2 z > C_{K,\mu} r + t \mid \mathcal{G} \right) \leq C_{p,K,\mu} \frac{r^{p/2} + r}{t^p} \quad \text{almost surely.}$$

Proof. Condition on $\mathcal{G}$. We reveal $Z_m, Z_{m-1}, \dots, Z_{m-r+1}$ in this order and iterate the recursion used in Lemma 3.1. For Lebesgue-a.e. z ,

$$\log^- |F_{m,r}(z)| \leq \sum_{q=1}^r \log^- \left| \frac{1}{z-Z_{m-q+1}} + C_q(z) \right|,$$

where $C_q(z)$ is $\mathcal{F}_{q-1}$ -measurable. Define

$$V_q := \int_K \log^- \left| \frac{1}{z-Z_{m-q+1}} + C_q(z) \right| d^2 z.$$

By Lemma 2.4 and Jensen's inequality,

$$\mathbb{E}[V_q \mid \mathcal{F}_{q-1}] \leq C_K, \quad \mathbb{E}[V_q^p \mid \mathcal{F}_{q-1}] \leq |K|^{p-1} \int_K \sup_{w \in \mathbb{C}} \mathbb{E} \left(\log^- \left| \frac{1}{z-Z_1} - w \right| \right)^p d^2 z \leq C_{p,K}$$

for some constant C_K and $C_{p,K}$. Therefore

$$\int_K \log^- |F_{m,r}(z)| d^2 z \leq \sum_{q=1}^r V_q, \quad \text{and} \quad \sum_{q=1}^r \mathbb{E}[V_q \mid \mathcal{F}_{q-1}] \leq C_K r.$$

So the centered sequence $(V_q - \mathbb{E}[V_q \mid \mathcal{F}_{q-1}])$ is a martingale difference sequence with uniformly bounded conditional variance and uniformly bounded conditional p -th moment. Therefore applying Lemma 3.3 gives

$$\mathbb{E} \left| \sum_{q=1}^r (V_q - \mathbb{E}[V_q \mid \mathcal{F}_{q-1}]) \right|^p \leq C_{p,K} (r^{p/2} + r).$$

Finally Markov's inequality gives the desired bound. Indeed,

$$\int_K \log^- |F_{m,r}(z)| d^2 z \leq \sum_{q=1}^r V_q \leq C_K r + \sum_{q=1}^r (V_q - \mathbb{E}[V_q \mid \mathcal{F}_{q-1}]).$$

Hence

$$\left\{ \int_K \log^- |F_{m,r}(z)| d^2 z > C_K r + t \right\} \subseteq \left\{ \left| \sum_{q=1}^r (V_q - \mathbb{E}[V_q \mid \mathcal{F}_{q-1}]) \right| > t \right\}.$$

□

In our main theorems, we do not require the whole root distribution μ to satisfy a logarithmic non-concentration condition. We only assume that μ contains a positive-mass regular part. The remaining part of μ is allowed to be arbitrary. It may contain atoms, singular components etc..

To separate out the portion of the sample on which we have estimates, we use a good/bad decomposition following the argument in the proof of [ANP26, Lemma 2.4]. Let $0 < \nu \leq \mu$ be the regular component. If $\nu(\mathbb{C}) = 1$, the bad component is absent; if $\nu(\mathbb{C}) < 1$, then let

$$\rho := \frac{\mu - \nu}{1 - \nu(\mathbb{C})}.$$

Then

$$\mu = \nu(\mathbb{C})\bar{\nu} + (1 - \nu(\mathbb{C}))\rho.$$

This means we can sample each root X_j as follows: with probability $\nu(\mathbb{C})$, sample X_j from $\bar{\nu}$. These are called *good roots*; with probability $1 - \nu(\mathbb{C})$, sample X_j from ρ . These are called *bad roots*. Recall here good means sampled from the nice logarithmically non-concentrated component $\bar{\nu}$, and bad means sampled from the leftover part ρ , on which we assume basically nothing.

The purpose of this decomposition is to isolate the randomness to which the lower-tail estimates apply. These estimates are available only for the good roots. After conditioning on the labels and on all bad roots, the bad roots become fixed and enter the elementary symmetric function only through z -dependent coefficients. Hence $e_k((z - X_1)^{-1}, \dots, (z - X_n)^{-1})$ becomes an inhomogeneous elementary symmetric function of the good reciprocal variables, with leading coefficient equal to one. This is exactly the form handled by the lower-tail lemmas.

Finally, since $\nu(\mathbb{C}) > 0$, the number of good roots is typically of order $\nu(\mathbb{C})n$. Since $k_n = o(n)$, with overwhelming probability there are at least k_n good roots. Thus the lower-tail recursion can be run using only the good roots, where the bad roots are absorbed into the coefficients.

Explicitly speaking, we can let (η_j) be i.i.d. Bernoulli random variables with $\mathbb{P}(\eta_j = 1) = \nu(\mathbb{C})$. If $\eta_j = 1$, sample X_j from $\bar{\nu}$; if $\eta_j = 0$, sample X_j from ρ . This gives the same marginal law μ . Since $M_n := |\{j \leq n : \eta_j = 1\}| \sim \text{Bin}(\nu(\mathbb{C}))$ and $k_n = o(n)$,

$$\mathbb{P}(M_n < k_n) \leq e^{-cn} \tag{3.1}$$

for all sufficiently large n and some $c > 0$. On the event $M_n \geq k$, condition on the labels (η_j) and on all bad roots. For Lebesgue-a.e. z , the full elementary symmetric function decomposes as

$$e_k \left(\frac{1}{z - X_1}, \dots, \frac{1}{z - X_n} \right) = \sum_{j=0}^k b_j(z) e_{k-j} \left(\frac{1}{z - Z_1}, \dots, \frac{1}{z - Z_{M_n}} \right), \tag{3.2}$$

where $Z_1, \dots, Z_{M_n}$ are the good roots, and $b_j(z)$ is the j th elementary symmetric function of the bad reciprocal variables. Thus Propositions 3.2 and Proposition 3.4 apply conditionally on the bad roots.

Now we have everything we need to prove the main theorems.

4. PROOFS OF THE THEOREMS

4.1. Proof of the convergence in probability theorem.

Proof of Theorem 1.4. The discrete case can be found in the first part of [ANP26, Theorem 1.3]. Let $0 < \nu \leq \mu$ be a compact component with

$$\sup_{w \in \mathbb{C}} \int_{\mathbb{C}} \log^- |x - w| d\bar{\nu}(x) < \infty,$$

and use the decomposition above. Fix a compact set $K \subset \mathbb{C}$. On $\{M_n \geq k\}$, (3.2) and Proposition 3.2 give

$$\mathbb{E} \left[\mathbf{1}_{\{M_n \geq k\}} \int_K \log^- |R_{n,k}(z)| d^2 z \right] \leq |K| \log \binom{n}{k} + C_K k.$$

Now use

$$\log \binom{n}{k} \leq k \log \frac{en}{k},$$

and since $k = o(n)$,

$$\frac{1}{n} \left(|K| \log \binom{n}{k} + C_K k \right) = o(1).$$

By Markov's inequality, since $\mathbb{P}(M_n < k) = o(1)$,

$$\frac{1}{n} \int_K \log^- |R_{n,k}(z)| d^2 z = o(1) \quad \text{in probability.}$$

Together with the deterministic positive part estimate in Lemma 2.3, we obtain

$$\frac{1}{n} \int_K \log |R_{n,k}(z)| d^2 z = o(1) \quad \text{in probability.}$$

Choose $\phi \in C_c^\infty(\mathbb{C})$, and compact K containing $\text{supp } \Delta\phi$. By Lemma 2.2,

$$\frac{1}{n} \sum_{\ell=1}^{n-k} \phi(W_\ell^{(n,k)}) - \frac{1}{n} \sum_{j=1}^n \phi(X_j) = o(1) \quad \text{in probability.}$$

And since

$$\left| \frac{1}{n-k} \sum_{\ell=1}^{n-k} \phi(W_\ell^{(n,k)}) - \frac{1}{n} \sum_{\ell=1}^{n-k} \phi(W_\ell^{(n,k)}) \right| \leq \frac{k}{n} \|\phi\|_\infty,$$

we get

$$\int \phi d\mu_{n,k} - \int \phi d\mu_n = o(1) \quad \text{in probability for every } \phi \in C_c^\infty(\mathbb{C}).$$

Here we've only showed convergence against compactly supported smooth test functions, we hope to pass it to every bounded continuous functions. So we still need to show that $\mu_{n,k}$ is tight, meaning with high probability most of its mass remains inside some fixed compact set.

Fix $\varepsilon > 0$. Choose $\chi \in C_c^\infty(\mathbb{C})$ such that $0 \leq \chi \leq 1$ and

$$\int \chi d\mu > 1 - \varepsilon/4.$$

Then $\int \chi d\mu_n \rightarrow \int \chi d\mu$ almost surely. Also above we showed that $\int \chi d\mu_{n,k} \rightarrow \int \chi d\mu_n$ in probability. Therefore, with probability tending to 1,

$$\mu_{n,k}(\text{supp } \chi) \geq \int \chi d\mu_{n,k} > 1 - \varepsilon.$$

Thus $(\mu_{n,k})$ is tight in probability. Together these imply

$$\mu_{n,k} \Rightarrow \mu \quad \text{in probability.}$$

□

4.2. Proof of the almost surely convergence theorem.

Proof of Theorem 1.5. Assume that μ has a positive compact L^p -logarithmically non-concentrated component for some $p > 2$. Fix a compact set $K \subset \mathbb{C}$. By (3.1),

$$\sum_{n=1}^{\infty} \mathbb{P}(M_n < k_n) < \infty.$$

On $\{M_n \geq k\}$, Proposition 3.4 and (3.2) imply that for every $t > 0$,

$$\mathbb{P}\left(\int_K \log^- |R_{n,k}(z)| d^2 z > |K| \log \binom{n}{k} + C_K k + t, M_n \geq k\right) \leq C_{p,K} \frac{k^{p/2} + k}{t^p}.$$

Let $\varepsilon > 0$. Since $k = o(n)$,

$$|K| \log \binom{n}{k} + C_K k = o(n).$$

For all sufficiently large n , this deterministic term is at most $\varepsilon n/2$. Taking $t = \varepsilon n/2$ and since $k \leq n$,

$$\mathbb{P}\left(\frac{1}{n} \int_K \log^- |R_{n,k}(z)| d^2 z > \varepsilon\right) \leq e^{-cn} + C_{p,K,\varepsilon} n^{-p/2}$$

for all sufficiently large n . Since $p > 2$, the right-hand side is summable. Borel–Cantelli Lemma yields

$$\frac{1}{n} \int_K \log^- |R_{n,k}(z)| d^2 z = o(1) \quad \text{almost surely for every compact } K.$$

Together with the deterministic positive part estimate in Lemma 2.3, we obtain

$$\frac{1}{n} \int_K |\log |R_{n,k}(z)|| d^2 z = o(1) \quad \text{almost surely for every compact } K.$$

Choose $\phi \in C_c^\infty(\mathbb{C})$, and compact K containing $\text{supp } \Delta\phi$. By Lemma 2.2,

$$\frac{1}{n} \sum_{\ell=1}^{n-k} \phi(W_\ell^{(n,k)}) - \frac{1}{n} \sum_{j=1}^n \phi(X_j) = o(1) \quad \text{almost surely.}$$

Then by the similar normalization change from $1/n$ to $1/(n-k)$ and a same tightness argument as in the proof of Theorem 1.4, we obtain

$$\mu_{n,k} \Rightarrow \mu \quad \text{almost surely.}$$

□

5. PROOF OF COROLLARY 1.6

Proof. We prove the corollary in steps.

Step 1: the discrete case. This follows directly from the discrete part of Theorem 1.5.

Step 2: a Frostman component gives the logarithmic hypothesis. Assume now that μ has a positive compact Frostman component. Thus there exist a compactly supported measure $0 < \nu \leq \mu$, $C < \infty$, and $s > 0$ such that

$$\nu(D(a, r)) \leq Cr^s, \quad a \in \mathbb{C}, \quad 0 < r \leq 1.$$

Let $\bar{\nu} := \frac{\nu}{\nu(\mathbb{C})}$. Then, for a possibly different constant C' ,

$$\sup_{a \in \mathbb{C}} \bar{\nu}(D(a, r)) \leq C' r^s, \quad 0 < r \leq 1.$$

We now record the Dini-type consequence of this estimate. Fix $p \geq 1$ and $a \in \mathbb{C}$. We have

$$\int (\log^- |x - a|)^p d\bar{\nu}(x) = p \int_0^\infty t^{p-1} \mathbb{P}(\log^- |X - a| > t) dt = p \int_0^\infty t^{p-1} \bar{\nu}(D(a, e^{-t})) dt.$$

Changing variables $r = e^{-t}$ gives

$$\int (\log^- |x - a|)^p d\bar{\nu}(x) = p \int_0^1 \left(\log \frac{1}{r}\right)^{p-1} \frac{\bar{\nu}(D(a, r))}{r} dr.$$

Taking the supremum over $a \in \mathbb{C}$, we obtain the Dini- p bound. Using the Frostman estimate,

$$\sup_{w \in \mathbb{C}} \int_{\mathbb{C}} (\log^- |x - w|)^p d\bar{\nu}(x) \leq C_p \int_0^1 \left(\log \frac{1}{r} \right)^{p-1} r^{s-1} dr < \infty.$$

Therefore μ has a positive compact L^p -logarithmically non-concentrated component for some $p > 2$. Theorem 1.5 gives $\mu_{n,k_n} \Rightarrow \mu$ almost surely whenever $k_n = o(n)$.

Step 3: the Doeblin case. If μ satisfies the Doeblin condition, then μ dominates a positive multiple of planar Lebesgue measure on some disk. Restricting to a smaller closed disk if necessary, we obtain a positive compact measure $0 < \nu \leq \mu$ such that

$$\nu(D(a, r)) \leq Cr^2, \quad a \in \mathbb{C}, \quad 0 < r \leq 1.$$

Thus ν is a positive compact Frostman component of μ . The conclusion follows from Step 2. This completes Step 3.

Step 4: the dimension-nondegenerate case. Suppose μ is dimension-nondegenerate in the sense of [ANP26]. Then there exist a finite measure $0 < \nu \leq \mu$ and a Borel set E with $\nu(E) > 0$ such that the lower local dimension of ν is positive at every point of E .

Hence, for some $s > 0$, the set

$$E_s := \{x \in E : \underline{\dim}_\nu(x) > 2s\}$$

has positive ν -mass. By the definition of lower local dimension and a countable-union argument, there exist a Borel set $F \subset E_s$ with $\nu(F) > 0$ and constants $C < \infty$, $r_0 > 0$ such that

$$\nu(D(x, r)) \leq Cr^s, \quad x \in F, \quad 0 < r < r_0.$$

By inner regularity, choose a compact set $F_0 \subset F$ with $\nu(F_0) > 0$, and set $\nu_0 := \nu|_{F_0}$. We claim that ν_0 is a Frostman component. If $D(a, r) \cap F_0 = \emptyset$, then $\nu_0(D(a, r)) = 0$. Otherwise, choose $x \in D(a, r) \cap F_0$. For $0 < r < r_0/2$, $D(a, r) \cap F_0 \subset D(x, 2r)$, and therefore

$$\nu_0(D(a, r)) \leq \nu(D(x, 2r)) \leq C'r^s.$$

For $r_0/2 \leq r \leq 1$, the same estimate is absorbed by increasing the constant. Thus

$$\nu_0(D(a, r)) \leq C''r^s, \quad a \in \mathbb{C}, \quad 0 < r \leq 1.$$

So ν_0 is a positive compact Frostman component of μ . The conclusion again follows from Step 2. $\square$

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